

## TAIL BOUNDS

Bernstein, Chernoff, and Hoeffding



$$\Pr \{ \sum X_i \geq t \} \leq \frac{\prod \mathbb{E}[e^{\lambda X_i}]}{e^{\lambda t}}$$

$$\Pr \{ \sum X_i \geq t \} \leq e^{-\frac{t^2}{2n}}$$

$$\Pr \{ \sum X_i \geq t \} \leq e^{-\mu \left( \frac{e\mu}{t} \right)^t}$$

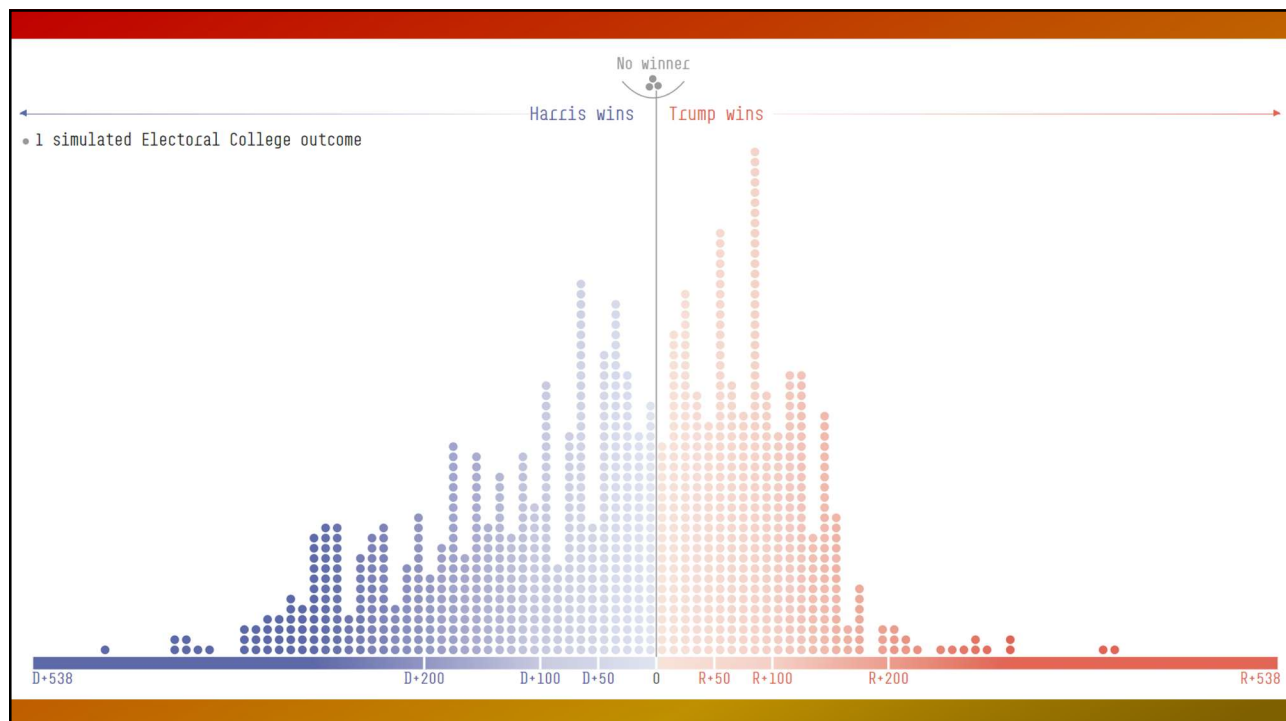
## Tail Bounds

Let  $X_1, X_2, \dots, X_n$  be independent identically distributed (i.i.d) random variables with Bernoulli/Rademacher distribution.

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound

$$\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$$



## Election Forecast

Two parties:  $-1$  и  $+1$ . We select  $n$  voters uniformly and independently. Ask them who they are going to vote for. Their responses:  $X_1, \dots, X_n$ .

### Outcome Prediction Algorithm

- Compute the average  $\bar{X} = \frac{1}{n} \sum X_i$
- If
  - $\bar{X} \leq -\delta$ , then we predict “-1”;
  - $\bar{X} \geq +\delta$ , then we predict “+1”;
  - $|\bar{X}| \leq \delta$ , then we output no prediction.

## Tail Bounds

Let  $X_1, X_2, \dots, X_n$  be independent identically distributed (i.i.d) random variables with Bernoulli/Rademacher distribution.

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound

$$\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$$

## Markov's Inequality

$X_1, X_2, \dots, X_n$  are i.i.d. random variables

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound  $\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$

Markov's inequality:

$$\Pr\{X_1 + \dots + X_n \geq t\} \leq \frac{n}{n + t}$$

$$\Pr\{X_1 + \dots + X_n \geq \delta n\} \leq \frac{1}{1 + \delta}$$

## Chebyshev's Inequality

$X_1, X_2, \dots, X_n$  are i.i.d. random variables

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound  $\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$

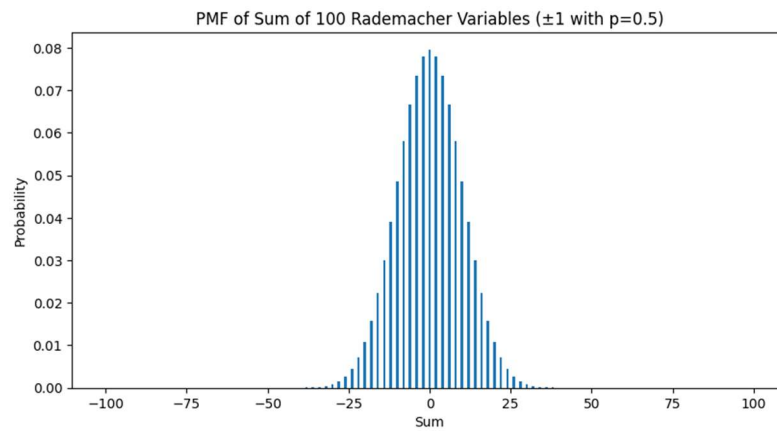
Chebyshev's inequality:

$$\Pr\{X_1 + \dots + X_n \geq t\} \leq \frac{\text{Var}[\sum X_i]}{t^2} = \frac{n}{t^2}$$

$$\Pr\{X_1 + \dots + X_n \geq \delta n\} \leq \frac{1}{\delta^2 n}$$

# Central Limit Theorem

De Moivre – Laplace theorem



# Central Limit Theorem

$X_1, X_2, \dots, X_n$  are i.i.d. random variables

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound  $\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$

Central Limit Theorem (CLT):

$$\Pr\{X_1 + \dots + X_n \geq t\} \lesssim \sqrt{\frac{n}{2\pi t^2}} e^{-\frac{t^2}{2n}} \quad ?$$

## Central Limit Theorem

$X_1, X_2, \dots, X_n$  are i.i.d. random variables

$$\Pr\{X_i = -1\} = \Pr\{X_i = 1\} = 1/2.$$

How can we bound  $\Pr\{X_1 + \dots + X_n \geq t\} \leq ?$

Central Limit Theorem:

$$\Pr\{X_1 + \dots + X_n \geq t\} \lesssim e^{-\frac{t^2}{2n}}$$

$$\Pr\{X_1 + \dots + X_n \geq \delta n\} \lesssim e^{-\frac{\delta^2 n}{2}}$$

## Bernstein Trick

Consider independent random variables

$$X_1, X_2, \dots, X_n$$

Fix a  $\lambda > 0$ . Let

$$f(x) = e^{\lambda x}$$

Then,  $f(x)$  is monotonically increasing function. Thus:

$$\{S \geq t\} = \{f(S) \geq f(t)\}$$

For ever r.v.  $S$ ; in particular, for  $S = \sum X_i$ .

## Bernstein Trick

$X_1, X_2, \dots, X_n$  — independent random variables.

Denote:  $S = \sum X_i$ . Let  $f(x) = e^{\lambda x}$ , where  $\lambda > 0$ .

Function  $f(x)$  is monotonically increasing function.

$$\{S \geq t\} = \{f(S) \geq f(t)\}$$

By Markov's inequality:

$$\Pr\{S \geq t\} = \Pr\{f(S) \geq f(t)\} \leq \frac{\mathbb{E}[f(S)]}{f(t)}$$

## Bernstein Trick

$X_1, X_2, \dots, X_n$  — independent random variables.

Denote:  $S = \sum X_i$ . Let  $f(x) = e^{\lambda x}$ , where  $\lambda > 0$ .

By Markov's inequality:

$$\Pr\{S \geq t\} \leq \frac{\mathbb{E}[f(S)]}{f(t)} = \frac{\mathbb{E}[e^{\sum \lambda X_i}]}{e^{\lambda t}} = \frac{\mathbb{E}[\prod e^{\lambda X_i}]}{e^{\lambda t}}$$

Since  $X_1, X_2, \dots, X_n$  are independent, we have

$$\Pr\{S \geq t\} \leq \frac{\prod \mathbb{E}[e^{\lambda X_i}]}{e^{\lambda t}}$$



## Bernstein Trick

$X_1, X_2, \dots, X_n$  — independent random variables.

For all  $\lambda > 0$ , we have

$$\Pr\{S \geq t\} \leq \frac{\prod \mathbb{E}[e^{\lambda X_i}]}{e^{\lambda t}}$$

## Hoeffding's Inequality

$X_1, \dots, X_n$  — i.i.d. random variables taking values  $\pm 1$  with probability  $1/2$ .

Then, for every positive  $t \geq 0$ :

$$\Pr\left\{\sum X_i \geq t\right\} \leq e^{-\frac{t^2}{2n}}$$

## Hoeffding's Inequality

$X_1, \dots, X_n$  – i.i.d. random variables taking values  $\pm 1$  with probability  $1/2$ . Then, for every positive  $t \geq 0$ :

$$\Pr\left\{\sum X_i \geq t\right\} \leq e^{-\frac{t^2}{2n}}$$

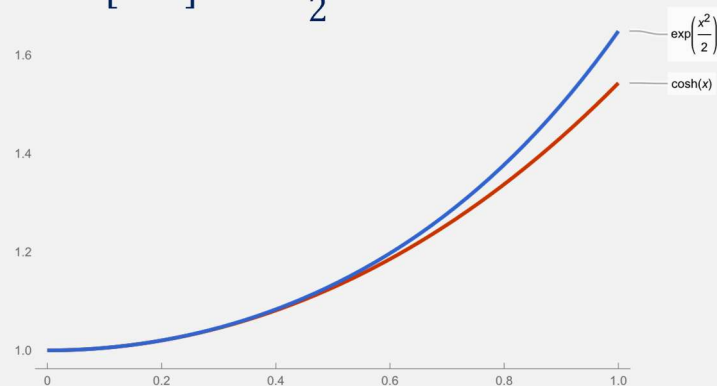
**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] = \frac{e^{\lambda} + e^{-\lambda}}{2} = \cosh \lambda$$

## Hoeffding's Inequality

**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] = \frac{e^{\lambda} + e^{-\lambda}}{2} = \cosh \lambda \leq e^{\lambda^2/2}$$



Plot: Wolfram Mathematica

## Hoeffding's Inequality

**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] = \frac{e^{\lambda} + e^{-\lambda}}{2} = \cosh \lambda \leq e^{\lambda^2/2}$$

Using Bernstein's lemma:

$$\Pr\{S \geq t\} \leq \frac{\prod e^{\lambda^2/2}}{e^{\lambda t}} = e^{n\lambda^2/2 - \lambda t}$$

For  $\lambda = t/n$ :

$$\Pr\{S \geq t\} \leq e^{-\frac{t^2}{2n}}$$

## Hoeffding's Inequality

$X_1, \dots, X_n$  – i.i.d. random variables taking values  $\pm 1$  with probability  $1/2$ . Then, for every positive  $t \geq 0$ :

$$\Pr\left\{\left|\sum X_i\right| \geq t\right\} \leq 2e^{-\frac{t^2}{2n}}$$

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

Then:

$$\Pr \left\{ \sum X_i \geq t \right\} \leq e^{-\mu} \left( \frac{e\mu}{t} \right)^t,$$

where  $\mu = \sum_i \mu_i$ .

**Remark:** This bound does not depend on  $n$ .

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

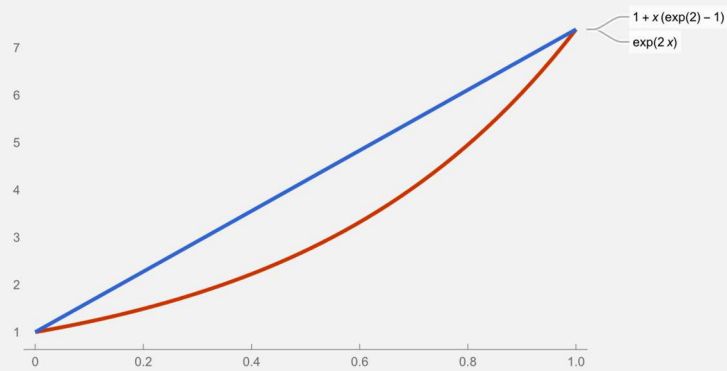
Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] \leq \mathbb{E}[1 + X_i(e^\lambda - 1)]$$

## Chernoff Bound

**Proof:**



$$e^{\lambda x} \leq 1 + x(e^{\lambda} - 1) \text{ for } x \in [0, 1]$$

Plot: Wolfram Mathematica

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0, 1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

**Proof:**

$$\begin{aligned} \mathbb{E}[e^{\lambda X_i}] &\leq \mathbb{E}[1 + X_i(e^{\lambda} - 1)] \\ &= 1 + \mu_i(e^{\lambda} - 1) \\ &\leq e^{\mu_i(e^{\lambda} - 1)} \end{aligned}$$

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] \leq e^{\mu_i(e^\lambda - 1)}$$

By Bernstein's lemma:

$$\Pr\{S \geq t\} \leq \frac{\prod e^{\mu_i(e^\lambda - 1)}}{e^{\lambda t}} = \frac{e^{\sum \mu_i(e^\lambda - 1)}}{e^{\lambda t}}$$

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

**Proof:**

$$\mathbb{E}[e^{\lambda X_i}] \leq e^{\mu_i(e^\lambda - 1)}$$

By Bernstein's lemma:

$$\Pr\{S \geq t\} \leq \frac{e^{\sum \mu_i(e^\lambda - 1)}}{e^{\lambda t}} = \frac{e^{\mu(e^\lambda - 1)}}{e^{\lambda t}}$$

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

Then:

$$\Pr \left\{ \sum X_i \geq t \right\} \leq \frac{e^{\mu(e^\lambda - 1)}}{e^{\lambda t}},$$

where  $\mu = \sum_i \mu_i$ . Let  $\lambda = \ln t/\mu$ :

$$\Pr \left\{ \sum X_i \geq t \right\} \leq \frac{e^{\mu(t/\mu - 1)}}{(t/\mu)^t}$$

## Chernoff Bound

$X_1, \dots, X_n$  – independent random variables.

Suppose: each  $X_i \in [0,1]$ . Let  $\mu_i = \mathbb{E} X_i$ .

Then, for  $\mu = \sum_i \mu_i$ :

$$\Pr \left\{ \sum X_i \geq t \right\} \leq \frac{e^{t-\mu}}{(t/\mu)^t} = e^{-\mu} \left( \frac{e\mu}{t} \right)^t$$

# TAIL BOUNDS

## Bernstein, Chernoff, and Hoeffding

